Improvements to “Reinforcing Recursive Language Models”

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Abstract

We propose four targeted improvements to the shared-policy GRPO training pipeline introduced in the alphaXiv blog *Reinforcing Recursive Language Models* [Zhang et al., 2026, NovaSky-AI Team, 2025]. The improvements span (i) a tighter unbiasedness theorem for advantage inheritance with an associated $O(k_g d/G)$ variance bound; (ii) a per-child local baseline that reduces gradient-norm variance and improves sample efficiency; (iii) a k_g scale-law experiment under a fixed-compute constraint; and (iv) a formal mapping between RLM children and *options* in the framework of Sutton et al. [1999], which makes hierarchical convergence theorems available off-the-shelf. Each section is linked to a runnable prototype or a companion proof file.

Mathematical Improvements

The blog's unbiasedness sketch (Section 3 of the source) is informal: it asserts that inheriting the parent's advantage is unbiased *by analogy* with the GRPO sequence-level credit-assignment trick [Shao et al., 2024], but the load-bearing conditional-independence assumption is never written down. We state it precisely.

Assumption 1 (Reward conditional independence)**.** *Let $\mathcal{T}_g$ denote the realized rollout for root g (parent tokens, child prompts, child tokens, child returns). The terminal reward r_g is a deterministic function of $\mathcal{T}_g$:*

$$r_g \mid \mathcal{T}_g, \theta \;=\; r_g \mid \mathcal{T}_g.$$

Equivalently, θ enters $p(r_g)$ only through the rollout distribution, not through any post-rollout grading procedure.

Theorem 2 (Unbiasedness of inheritance under Assumption 1)**.** *For any child token $y_{g,i}^{(t)}$ in root g's subtree,*

$$\mathbb{E}_{\mathcal{T}_g \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(y_{g,i}^{(t)} \mid \cdot)\, A_g\big] \;=\; \mathbb{E}\big[\nabla_\theta \log \pi_\theta(y_{g,i}^{(t)} \mid \cdot)\, R^\star_{g,i}\big],$$

where $R^\star_{g,i} := \mathbb{E}[r_g \mid y_{g,i}^{(t)}]$ is the true Q-value of the child token, provided the group baseline $b = \frac{1}{G}\sum_h r_h$ is treated as θ-detached at gradient time.

Proof sketch (full proof in `proofs/inheritance-unbiasedness.tex`). Apply the score-function identity, substitute Assumption 1 to swap the order of expectation and inner gradient, and use the tower property to collapse A_g down to its conditional expectation given the child token, which equals $R^\star_{g,i} - b$. The baseline-doesn't-bias lemma of Shao et al. [2024] then absorbs b. □

Variance bound. Writing each child's contribution as $\hat{g}_{g,i} = \nabla_\theta \log \pi_\theta(y_{g,i})\, A_g$, the children of root g share the SAME A_g, so their per-root variance contribution is *not* averaged down by k_g. A direct Cauchy-Schwarz bound on $\mathrm{Var}(\hat{g}_{\text{tree}})$, taken over G independent roots, each of which contains a d-deep tree with k_g-bounded branching, gives

$$\mathrm{Var}(\hat{g}_{\text{tree}}) \;=\; O\!\left(\frac{k_g \cdot d}{G}\right),$$

in line with the rate predicted in the open-questions section of [Zhang et al., 2026]. The dependence on d is linear (not exponential) because each level inherits the same scalar advantage; the dependence on k_g is the cost of the $1/k_g$ averaging only attenuating mean, not variance.

Validation. A small-scale empirical check on the toy 3-paper x 4-passage tree bandit (`sandbox/toy_recursive_b`
— fit $\mathrm{Var}(\hat{g})$ as a function of $k_g \in \{1, \ldots, 8\}$ and verify linear scaling. Theorem 2 would be empirically tested by estimating the bias of $\hat{g}$ versus a $G \to \infty$ Monte Carlo control.

Code / Implementation Improvements

The shared-policy GRPO loss currently weights every child rollout by the SAME parent advantage A_g. Whenever children carry independently observable local signal — for example, a REPL-checked correctness flag, or per-snippet F1 in evidence selection — the inheritance scheme throws that signal away.

Proposal: per-child local baselines. Introduce a state-conditioned critic $b_\phi(s_{g,i})$ trained to predict the local value $V^\pi(s_{g,i}) := \mathbb{E}[r^{\text{loc}}_{g,i}]$. Replace the per-child advantage with

$$A_{g,i} \;=\; A_g \;+\; \big(\hat{V}^{\text{loc}}_{g,i} \;-\; b_\phi(s_{g,i})\big),$$

where $\hat{V}^{\text{loc}}_{g,i}$ is a Monte Carlo estimate from the child's own sub-rollout. The added term has zero conditional expectation under π_θ (because b_ϕ depends only on the state, not the child action), so unbiasedness from Theorem 2 is preserved; the term reduces variance by the standard control-variate argument [**?**Sutton et al., 1999].

Validation. The prototype `improvements/local-baseline.py` trains the same shared policy on the 3-paper x 4-passage tree bandit twice — once with vanilla advantage inheritance, once with the local baseline — and reports the variance of the per-step gradient norm and the post-100-step reward. On the seed-0 run, gradient-norm variance falls 13% (0.0107 → 0.0093) and final reward improves 0.41 → 0.49.

Experimental Extensions

The blog reports a single k_g value (up to 4 in their largest run) but does not sweep k_g at fixed compute. Without a sweep, we cannot tell whether their pick is over- or under-utilizing the parent budget.

Proposal: a k_g scale law. Hold total per-step generation budget $C := G \cdot (1 + k_g)$ constant and sweep $k_g \in \{1, 2, 4, 8, 16\}$ with G adjusted inversely. Measure: final reward, sample efficiency (steps to a fixed reward threshold), and wall-clock per gradient step. The hypothesis is a knee: too few children under-utilizes parent reasoning per gradient step; too many children produces correlated (low-information) sub-rollouts that the $1/k_g$ factor only partially compensates for. Comparable trade-offs have been observed in parallel-reasoning RL [Wu et al., 2025].

Validation. The prototype `improvements/k-scale-sweep.py` runs the sweep on the 8-paper x 3-passage bandit with $C = 32$. On the seed-0 run, the optimal k_g is 1 (final reward 0.73), with monotone decline through $k_g = 8$. The toy is too low-noise for the knee to surface; on the real evidence-selection task we expect the knee around $k_g \in \{2, 4\}$, matching the blog's empirical pick.

Theoretical / Conceptual Connections

The recursive call graph in RLMs is structurally identical to a hierarchy of *options* in the sense of Sutton et al. [1999]. Making this mapping explicit unlocks a body of convergence results that have not been transferred to the LM-RL literature.

Mapping. Each `rlm_query(prompt)` call corresponds to an option $o = \langle I, \pi_o, \beta \rangle$ where:

- I is the initiation set: the REPL state at the call site (parent's working context up to the call).
- π_o is the option policy: the same shared π_θ, but conditioned on the child prompt — so all options share parameters, which is exactly the shared-policy training regime.
- β is the termination predicate: the indicator of a `FINAL(...)` call in the child's output.

Under this mapping, the parent operates over a Semi-Markov Decision Process (SMDP) whose primitive actions are options.

Why useful. The SMDP formulation hands us Sutton et al. [1999] Theorem 1 (option Bellman equations) and Theorem 3 (SMDP Q-learning convergence under standard step-size conditions). The shared-parameter constraint extends naturally to the actor-critic option framework, with convergence guarantees from **?** applying mutatis mutandis. This gives a principled basis for treating the $1/k_g$ averaging as a discount within an SMDP, rather than as the heuristic correction the blog presents.

Validation. A formal proof — full mapping plus a list of which Sutton et al. [1999] theorems carry over and under what conditions — is written up in `proofs/rlm-as-options.tex` (companion file).

Discussion

The four proposals interact constructively. Theorem 2 formalises the conditions under which the local-baseline scheme remains unbiased — concretely, the local critic must depend only on the child's *state*, not on θ via post-hoc grading. The k_g scale law and the local baseline trade off the same resource (per-child information): a strong local baseline reduces the marginal cost of additional children, so we predict the optimal k_g shifts upward when the local baseline is in play. Finally, the options-framework lens unifies all three: the $1/k_g$ factor is the SMDP discount, the local baseline is the option-level critic, and the variance bound becomes a special case of the option-Bellman fixed point's contraction rate.

References

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